MILITARY SPECIFICATION

SWITCHING UNIT
SA-2498/ARC

This specification is approved for use within the Naval Air Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the design, performance, and acceptance requirements for Switching Unit SA-2498/ARC hereinafter referred to as switching unit.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Systems Engineering and Standardization Department (SESD) Code 53, Lakehurst, NJ 08733-5100 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

AMSC N/A

FSC 5821

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

Federal

QQ-P-416	Plating, Cadium (Electrodeposited)

Military

MIL-P-116	Preservation-Packaging, Methods of
MIL-C-172	Cases; Bases, Mounting; and Mounts, Vibration
MIL-B-5087	Bonding, Electrical, and Lightning Protection, for Aerospace Systems
MIL-W-5088	Wiring; Aerospace Vehicle
MIL-E-5400	Electronic Equipment, Airborne, General Specification for
MIL-T-5422	Testing, Environmental, Aircraft Electronic Equipment
MIL-C-5541	Chemical Conversion Coating on Aluminum and Aluminum Alloys
MIL-E-17555	Electronic and Electrical Equipment, Accessories, and Repair Parts, Packaging and Packing of
MIL-T-18303	Test Procedures; Preproduction, Acceptance and Life for Aircraft Electronic Equipment, Format for
MIL-N-18307	Nomenclature and Identification for Electronic, Aeronautical, and Aeronautical Support Equipment Including Ground Support Equipment
MIL-S-19500	Semiconductor Devices, General Specification for
MIL-M-38510	Microcircuits General Specification for
MIL-C-38999	Connectors, Electrical, Circular, Miniature, High Density, Quick Disconnect, Environmental Resistant, Removable Crimp Contacts; General Specification for
MIL-C-39018	Capacitor, Fixed Electrolytic (Aluminum Oxide), General Specification for
MIL-H-46855	Human Engineering Requirements for Military Systems, Equipment and Facilities

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

MIL-P-55110	Printed Wiring Boards
MIL-R-85664	Receiver-Transmitter, Radio RT-1250A/ARC
MIL-R-85665	Receiver-Transmitter, Radio RT-1324A/ARC

STANDARDS

Federal

FED-STD-595	Federal Standard Colors

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-275	Printed Wiring for Electronic Equipment
MIL-STD-415	Test Provisions for Electronic Systems and Associated Equipment, Design Criteria for
MIL-STD-454	Electronic Equipments, General Requirements for
MIL-STD-461	Electromagnetic Emission and Susceptibility Measurements for the Control of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-471	Maintainability Demonstration
MIL-STD-701	Lists of Standard Semiconductor Devices
MIL-STD-704	Aircraft Electric Power Characteristics
MIL-STD-756	Reliability Prediction
MIL-STD-781	Reliability Tests; Exponential Distribution
MIL-STD-785	Reliability Program for Systems and Equipment Development and Production
MIL-STD-794	Parts and Equipments, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods
MIL-STD-883	Test Methods and Procedures for Microelectronics
MIL-STD-965	Parts Control Program

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities
MIL-STD-1562	Lists of Standard Microcircuits
MIL-STD-2074	Failure Classification for Reliability Testing
MIL-STD-2076	Unit Under Test Compatibility with Automatic Test Equipment, General Requirements for
MIL-STD-2077	Test Program Sets, General Requirements for
MIL-STD-2084	Maintainability of Avionics and Electronic Systems and Equipment, General Requirements for

HANDBOOKS

Military

MIL-HDBK-217	Electronic Equipment, Reliability Prediction of
MIL-HDBK-235	Electromagnetic (Radiated) Environment Considerations for Design and Procurement of Electrical and Electronic Equipment, Subsystems, and Systems

2.1.2 Other Government documents. The following other Government documents, form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DOCUMENTS

Naval Air Systems Command

AD-1115	Electromagnetic Compatibility Design Guide for Avionics and Related Ground Support Equipment
WS-6536	Process Specification Procedures and Requirements for Preparations and Soldering of Electrical Connections

National Security Agency

CSEEB-13	TSEC/KY-28 Speech Security Equipment Engineering Bulletin
CSEEB-14	Communications Security Equipment System Document for TSEC/KY/57/58
CSEEB-32	TSEC/KY-57/58 Communications Security Equipment Engineering Bulletin

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

Department of the Navy

NAVMAT P-4855.1 Navy Power Supply Reliability

NAVMAT P-9492 Navy Manufacturing Screening Program

(Copies of specifications, standards, handbooks, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards) the text of this specification shall take precedence. Nothing in this specification however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3.0 REQUIREMENTS

3.1 Item description. The switching unit shall function exclusively as part of an equipment complement. When used in conjunction with its complement the switching unit enables operator controlled re-configuration capability. The equipment complement of the switching unit is shown in Table I.

Table I. Compatible equipment types

Device	Compatible Equipment Type
Auto-Relay Control Unit	C-10777 A/ARC C-11132/ARC C-11628/ARC
Radio Set	AN/ARC-182(V) AN/ARC-159(V)-1
COMSEC Encryption Device	TSEC/KY-58 TSEC/KY-28

3.2 First article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.4).

3.3 Parts. In the selection of parts, the prime considerations are fulfillment of performance requirements, logistics supportability and reliability assurance in a cost effective design. The following shall govern in parts selection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

a. Subassemblies shall be designed and fabricated to be repairable. Maximum economic standardization of parts and materials shall be exercised. Parts selection shall be made in the following order of precedence:

(1) Military Standard parts shall be selected in accordance with MIL-STD-454, Requirement 22.

(2) Nonstandard parts previously qualified for other programs.

(3) Nonstandard parts.

b. Nonstandard parts and nonestablished reliability military standard parts shall be screened to the requirements of the most similiar military standard or standard nonestablished reliability part. Integrated circuits and hybrids shall use MIL-STD-883, Method 5004, Class B, as a basis for tailored screens. Purchase specifications shall specifically delineate screening requirements.

3.3.1 Component rescreening. The contractor shall conduct 100 percent incoming inspection of all active components used in production equipment and spares. Functional and parametric checks of active components shall be performed at the components rated temperature extremes and at ambient. Components which are not compliant with their specification shall not be used in the Switching Unit.

3.3.2 Parts derating and application. All parts used shall be applied well within their ratings. The derating shall encompass the appropriate and meaningful application conditions such as voltage, current, power, temperature, mechanical, and duty cycle. Electronic and electromechanical parts shall conform to Table II electronic parts derating for worst case electrical and environmental stress unless formal written approval is received from the procuring activity prior to incorporation into the design. Part level stress analysis shall be used to verify that all parameter stresses are within the derated values at worst case circuit and environmental conditions. The following limitations on parts usage shall apply:

a. Prohibited application

(1) Non-hermetically sealed semiconductors except power transistors
(2) Photo couplers (photo transistors)
(3) Hot carrier (Schottky) rectifiers
(4) Microcircuit sockets
(5) Non-hermetically sealed wet tantalum capacitors

3.3.3 Parts control. The contractor shall establish and maintain a parts control program in accordance with the requirements of MIL-STD-965, Procedure I and as specified herein. The program shall control part standardization and the criteria for the selection, application, and testing of parts to achieve the reliability required. As a minimum, a parts program equivalent to that specified herein shall be required of subcontractors and suppliers.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

Table II. Electronic parts derating.

Part Type	Parameter	Maximum Allowable Application Stress
Capacitor	Applied Voltage (Ripple Voltage for Tantalum)	50% of rating 1/
Resistors	Power Voltage	50% of rating 70% of rating
Diodes	Power Current Voltage Junction Temperature	50% of rating 75% of rating 60% of rating 2/
Transistors	Power Voltage Junction Temperature	55% of rating 60% of rating 2/
Integrated Circuits		
Digital	Fan-Out, Fan-In Junction Temperature	80% of rating 2/
Linear	Power Junction Temperature	75% of rating 2/
Transformers, Inductive Devices	Parts shall be used at a winding temperature of at least 30 degrees below rated temperature. In addition, the maximum voltage between windings or between a winding and a shield shall be not greater than 60% of rated voltage for the steady state conditions and 90% of rated voltage with transients. General purpose inductors, audio and power transformers shall be applied at no more than 70% of rated current.	

1/ For solid tantalum and all electrolytic capacitors, the maximum applied voltage shall be not greater than 70 percent of the vendor rated value. "Applied voltage" is the sum of the applied peak ripple voltage and the applied DC voltage.

2/ Semiconductor devices shall be mounted and operated in such manner to insure worst-case junction temperature is not greater than 110 degrees C at +71 degrees C ambient temperature. Any deviation requires procuring activity approval. Power semiconductor devices, on a very selective basis, may be mounted and operated with worst case junction temperatures not greater than 150 degrees C at +71 degrees C ambient temperatures.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the current version before use.

3.3.4 Non-standard parts approval. The contractor shall obtain procuring activity approval of all non-standard parts used in the equipment. Standard parts are defined in MIL-E-5400. In addition to the MIL-E-5400 requirements for standard parts, microelectronic parts are standard only if:

a. The device is listed in MIL-STD-1562, and

b. Sources (at least two) are listed on the Qualified Parts List (QPL), and

c. The part is procured from one of the qualified sources, and

d. Quality level "B" (or better) of MIL-M-38510 is required.

The approval procedure shall be in accordance with MIL-E-5400 and MIL-STD-965, Procedure I. In addition to the requirements of MIL-E-5400, approval of non-standard parts will be based on:

a. Suitability for the application

b. Conformance to this specification and to MIL-E-5400

c. Interchangeability

d. Screening (burn-in) method

3.3.5 Samples of non-standard parts and materials. Samples of non-standard parts and materials are not required unless specified in the contract. Where non-standard parts or materials samples are specified in the contract, the sample shall be in accordance with MIL-E-5400 (see 6.2.1).

3.3.6 Cabling and connections.

3.3.6.1 Cables and connectors. The switching unit shall provide for the use of cables and connectors in accordance with MIL-E-5400. The design of the connectors shall be such that improper assembly, mating or installation is impossible. Non-RF connectors shall meet the requirements of MIL-C-38999. All external connectors shall be scoop-proof, have cadmium plating finish in accordance with QQ-P-416 (over a suitable underplating) to withstand a 500 hour salt test, and provide 360 degrees circumferential grounding techniques for grounding prior to contact engagement.

3.3.6.2 Interconnection cabling. The switching unit shall be capable of required operation using external wiring in accordance with the applicable requirements of MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provide the assembly of the cable to its plugs may be easily accomplished. External cables and that portion of the connectors attached to the cables shall not be supplied as part of the switching unit.

3.3.7 Interchangeability. Physical and functional interchangeability shall exist for all units and replaceable assemblies, subassemblies, and parts of the switching unit in accordance with MIL-E-5400. The switching unit shall

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the current version before use.

be constructed to preclude the requirement for any adjustment when replacing an item by another item of the same type designation.

3.4 Design and construction. The switching unit shall conform with all the applicable requirements of MIL-E-5400 for design and construction except where otherwise specified herein. In any case, it is a condition of final acceptance that the switching unit shall meet all the performance requirements of this specification.

3.4.1 Reliability. When specified in the contract or purchase order, the contractor shall establish a reliability program conforming to MIL-STD-785 (see 6.2.1). Task 104 of MIL-STD-785 shall be implemented to establish and maintain a closed-loop failure reporting system. This system will identify failure trends and patterns at all levels of assembly prior to acceptance of the hardware and will also identify failure trends related to field failures throughout the contract period. The contractor shall develop and implement corrective action for significant failure trends. The reliability program shall include provisions for assuring the reliability of the switching unit or any portion of the equipment when supplied for use as spare or repair part.

3.4.1.1 Operational stability. The switching unit shall operate with specified performance for at least the specified mean (operating) time between failures without the necessity for readjustment of any controls which are inaccessible to the operator during normal use.

3.4.1.2 Operating life. The switching unit shall have a total operating life of 20,000 hours with minimal servicing and replacement of parts. Parts requiring scheduled replacement shall be specified by the contractor.

3.4.1.3 Reliability in MTBF (mean time between failure). The switching unit shall have a specified mean (operating) time between failures, upper test MTBF (θ_0) of 12,000 hours (as specified in MIL-STD-781), and a lower test MTBF (θ_1) of 6,000 hours when tested and accepted as specified under the requirements of 4.4.5 and 4.5.4.

3.4.2 Maintainability. Maintainability construction and packaging provisions for test points and other maintainability parameters shall be as specified in MIL-STD-2084 except for built-in-test.

3.4.2.1 Compatibility with ATE. The switching unit shall be compatible with Automatic Test Equipment (ATE) for WRA testing and fault isolation to a faulty SRA in accordance with MIL-STD-2076.

3.4.2.2 Maintainability design. All electrical or electronic circuits and parts shall be packaged on replaceable and repairable, plug-in modules. Circuit breakers, indicators, and other items which may logically be bolted and soldered, or otherwise more rigidly electrically and mechanically fastened to the switching unit housing shall not be considered modules. SRAs (Shop Replacement Assemblies) shall be QRAs (Quick Replaceable Assemblies) as specified in MIL-STD-2076. SRA arrangement shall be such that access to any SRA does not require the removal of adjacent SRAs or parts other than access panels. The use of a special tool or tools shall not be required for SRA removal. Fuses may be used in lieu of circuit breakers.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the currentversion before use.

3.4.2.2.1 Keying. Connectors shall be keyed to prevent the insertion of a connector or subassembly into an improper location within or on the switching unit. All connectors shall be legibly marked with reference designations.

3.4.2.2.2 Access panels. Access panels shall be retained by fasteners suitable to the procuring activity.

3.4.2.2.3 Service access. Fasteners required to be loosened for removal of the WRA shall be held by retaining devices suitable to the procuring activity.

3.4.2.2.4 Adjustments. No adjustment or alignment shall be required at the organizational level. Adjustment or alignment if required at a higher maintenance level (intermediate or depot), shall not require the removal of the part to be adjusted or aligned, or of the hardware element to which it is attached.

3.4.2.2.5 Test points. All test points shall be in accordance with MIL-STD-2084 and MIL-STD-415 to accommodate approved test equipment. MIL-STD-2084 takes precedence over MIL-STD-415. The sub-SRA test points and related connector requirements of MIL-STD-2084 are waived where prohibited by space or circuit performance considerations. The sub-SRA test points and related connector requirements philosophy shall be reflected in appropriate maintenance data. Conformal coating may be applied to the PC pads used as test points.

3.4.2.2.5.1 Module test points. Module test points shall be accessible while power remains applied to the modules. Test points shall become accessible with minimum removal of protective covers or enclosures.

3.4.2.2.5.2 Test points utilized at intermediate or depot level. Test points shall be provided for verification of faulty equipments and isolation of faulty modules at the intermediate level and repair of equipments and modules at the depot level.

3.4.3 Transportability. The switching unit shall be transportable by air, rail, truck and ship when packaged as specified in Section 5.

3.4.4 Standard conditions. The following conditions shall be used to establish normal performance characteristics under standard conditions and for making laboratory bench tests, except that required field tests may be conducted under outside ambient conditions:

a.	Temperature	Room ambient
b.	Altitude	Normal ground
c.	Vibration	None
d.	Humidity	Room ambient up to 90 percent relative humidity
e.	Input power voltage	28 $\pm$ 0.5 VDC

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

3.4.4.1 Service conditions. The switching unit shall operate as required under any of the environmental service conditions or combination of these conditions as specified in MIL-E-5400 for Class 2 equipment, except as modified herein.

3.4.4.1.1 Sinusoidal vibration. The switching unit shall operate as required when subjected to sinusoidal vibration in accordance with MIL-T-5422 to the requirements of MIL-E-5400, Curve IIIB, when mounted on the MT-4609/ARC-159.

3.4.4.1.2 Random vibration. The switching unit shall operate as required when subjected to the random vibration requirements specified in 4.4.4.

3.4.4.1.3 Thermal conditions. The switching unit shall operate as required under the thermal conditions specified by MIL-E-5400, Class 2, without the requirement for auxiliary cooling.

3.4.4.1.4 Temperature shock. The switching unit shall meet the temperature shock requirements of MIL-STD-810, Method 503.2.

3.4.4.2 Warm-up time. The time required for the switching unit to warm up prior to operation shall be kept to a minimum and shall be not greater than 10 seconds at standard conditions. Under service conditions, the warm-up time shall be not greater than 20 seconds.

3.4.4.3 Electromagnetic interference (EMI). The switching unit shall comply with the requirements of MIL-STD-461 for Class A1b equipment except as modified by this specification. These requirements shall be met in all operating modes of the unit. MIL-STD-461, is modified by Tables III and IV and the following additions and deletions:

a. Add: NAVAIR AD-1115

b. Delete:

(1) AFSC DH 2-5
(2) AFSC DH 2-7
(3) AMC Pamphlet 706, 235
(4) NAVELEX 01001, 106

3.4.4.3.1 Conducted and radiated emissions limits. The conducted and radiated emissions of the switching unit shall comply with the limits of the following requirements of MIL-STD-461 Part 2:

a. CE01

b. CE03

c. CE07

d. RE01

e. RE02

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

3.4.4.3.2 Conducted and radiated susceptibility limits. The switching unit shall comply with conducted and radiated susceptibility limits of the following requirements of MIL-STD-461:

a. CS01

b. CS02

c. CS06

d. RS01

e. RS02

f. RS03

The switching unit shall exhibit no susceptibility as specified in 3.4.4.3.3 when subjected to an RS03 test to the limits of Table III. In addition, the switching unit shall exhibit no susceptibility in accordance with the criteria of 3.4.4.3.3b when subjected to signals at the levels of Table IV.

3.4.4.3.3 Susceptibility. The following criteria shall be applied to determine whether the switching unit is susceptible:

a. Any failure of the switching unit to comply with the requirements of 3.6.1.1 through 3.6.1.10 shall be considered an indication of susceptibility.

b. Any repeatable non-commanded change in the modes of operation of the switching unit shall be considered an indication of susceptibility. This shall apply whether the failure is transient (momentary) or non-transient (permanent) in nature. In addition the inability to change mode of operation while exposed to the susceptibility signal shall be considered an indication of susceptibility.

3.4.4.3.4 Ground plane interference. All switching unit interface circuits shall be capable of specified performance when subjected to the following aircraft chassis noise between the WRA's.

a. Three volts rms from 320 Hz to 500 Hz. (Not to be greater than 150ma rms applied current.)

b. One volt rms from 500 Hz to 20 MHz. (Not greater than 150ma rms applied current for frequencies between 500 Hz and 50 kHz. Not to be greater than 1 watt (CW) from a 50-ohm source for frequencies above 50 kHz.)

c. $\pm$ eight volt pulses, 100 usec wide at 100 pps. (Not to be greater than 15a peak current spikes.)

In addition to the susceptibility definitions of 3.4.4.3.3, the radio shall maintain intelligible communications during a "talk thru" test with another radio set using a low level signal (not to exceed 20 microvolts input to the receiver). The test will be performed with a test set up employing a RT-1250A/ARC, RT-1324A/ARC, or equal receiver-transmitter.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the current version before use.

Table III. RS03 signals

Frequency (MHz)	Modulation				Power Density (mW/cm^2) Peak 5/	Avg	v/m (ref) 6/
0.014-850 7/	AM 90% 1 kHz Sine Wave				-	.106	20
215-225	PAM 1/	20 usec 2/	PW 3/	200 pps 4/	.317	-	35
215-225	PAM	200 usec	PW	200 pps	.317	-	35
400-450	PAM	15 usec	PW	300 pps	.106	-	20
850-940	PAM	10 usec	PW	1000 pps	.106	-	20
850-940	PAM	125 usec	PW	300 pps	.106	-	20
1215-1365	PAM	10 usec	PW	3000 pps	.106	-	20
1365-2900	PAM	1 usec	PW	1000 pps	.106	-	20
2900-3100	PAM	3 usec	PW	1000 pps	1.6	-	78
2900-3100	PAM	70 usec	PW	200 pps	1.6	-	78
3100-9000	PAM	1 usec	PW	1000 pps	.106	-	20
9000-14000	PAM	0.2 usec	PW	1000 pps	.106	-	20
14000-18000	PAM	0.2 usec	PW	50000 pps	.106	-	20

1/ Pulse Amplitude Modulation.
2/ Microseconds.
3/ Pulse Width.
4/ Pulses Per Second.
5/ Peak Power Density for PAM signals is specified as the power density measured by an average power measuring device if the generator is run CW; average power density within the pulse envelope.
6/ Volts Per Meter equivalent shown for reference only.
7/ Frequency ranges listed with more than one modulation shall be repeated as necessary to include all different modulations and field strength shown.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:51Z
Check the source to verify that this is the current version before use.

Table IV. High level RS03 signals

Frequency (MHz)	Modulation				Power Density (mw/cm^2) 5/	v/m (ref) 6/	Scan Rate (sec) 7/	Dwell Time (msec) 7/
220 8/	PAM1/	10 usec2/	PW3/	200 pps4/	10.6	200	4	500
220	PAM	200 usec	PW	200 pps	10.6	200	4	500
425	PAM	15 usec	PW	300 pps	10.6	200	10	200
900	PAM	2 usec	PW	1000 pps	10.6	200	5	50
900	PAM	125 usec	PW	300 pps	10.6	200	5	50
1250	PAM	10 usec	PW	3000 pps	10.6	200	5	50
1350	PAM	5 usec	PW	3000 pps	10.6	200	5	50
3000	PAM	3 usec	PW	1000 pps	10.6	200	5	50
3000	PAM	70 usec	PW	200 pps	10.6	200	5	50
3600	PAM	1 usec	PW	1000 pps	10.6	200	5	50
5600	PAM	1 usec	PW	600 pps	10.6	200	5	50
9100	PAM	0.5 usec	PW	1000 pps	10.6	200	5	50
9375	PAM	0.2 usec	PW	1000 pps	10.6	200	5	50
13900	PAM	0.2 usec	PW	50000 pps	10.6	200	0	CONT
15500	PAM	0.2 usec	PW	50000 pps	10.6	200	0	CONT
33200	PAM	0.2 usec	PW	2000 pps	10.6	200	0	CONT

1/ Pulse Amplitude Modulation.
2/ Microseconds.
3/ Pulse Width.
4/ Pulses Per Second.
5/ Peak Power Density for PAM signals is specified as the power density measured by an average power measuring device if the generator is run CW; average power density within the pulse envelope.
6/ Volts Per Meter equivalent shown for reference only.
7/ Scan and Dwell refer to simulated antenna rotation. The signal shall be disabled except for the specified dwell. The signal shall be periodically enabled for the specified dwell time with the period equal to the specified scan time.
8/ Frequency ranges listed with more than one modulation shall be repeated as necessary to include all different modulations shown.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

3.4.4.3.5 Conducted susceptibility test, interconnecting and signal leads. The switching unit shall exhibit no susceptibility, as specified by 3.4.4.3.3 when subjected to 100 MV rms 2 to 30 MHz coupled into its interconnecting and signal leads.

3.4.4.3.6 Grounding requirements. The grounding system between WRA's of the equipment shall be compatible and consistent with the aircraft grounding scheme specified in 3.4.4.3.6.1 through 3.4.4.3.6.3. All WRA's shall be grounded in such a manner as to prevent ground loops and common ground returns for signal and power circuits, provide effective shielding for signal circuits, minimize EMI, and to protect personnel from electrical hazards.

3.4.4.3.6.1 Chassis grounds. A ground wire of minimum length connected internally to the WRA chassis shall be provided at a pin on each primary power connector. No circuit shall be allowed to utilize this wire as its primary return.

3.4.4.3.6.2 Shielding grounding. A separate connector pin shall be provided for each aircraft wire shield unless suitable grounding is provided in external cable backshells. The connector shield pins shall be grounded to the equipment chassis inside the WRA adjacent to the connector mounting and by the shortest means practical.

3.4.4.3.6.3 Component grounds. All externally exposed metal parts, shields, control shafts, switch handles, connectors, and bushings shall be grounded to the chassis.

3.4.4.3.7 Shielding gaskets. Shielding gaskets and similar devices for RF applications shall avoid the use of materials which represent severe dissimilar metal combinations with the interface.

3.4.5 Dimensions. The dimensions of the switching unit shall be as shown on Figure 1.

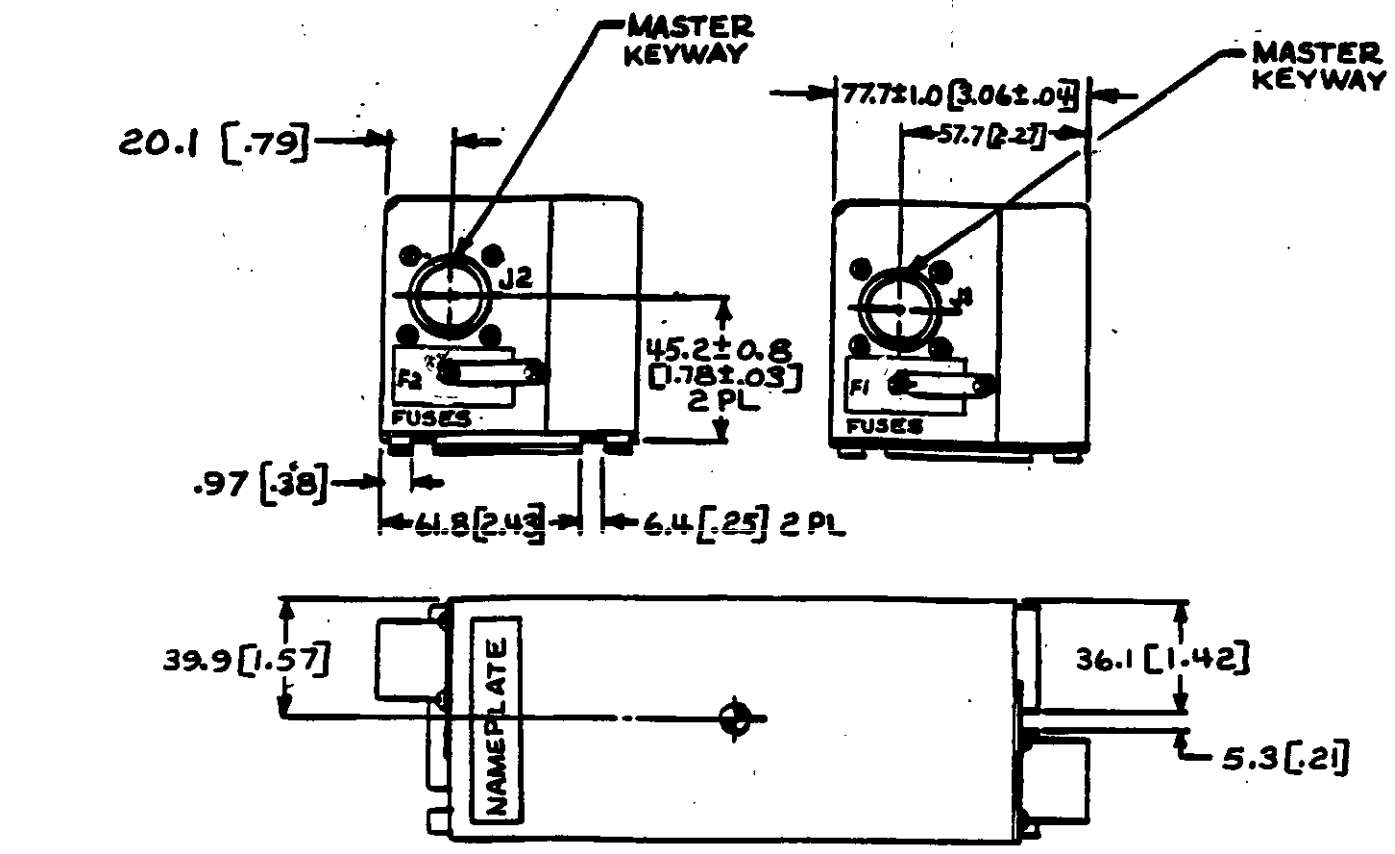
3.4.6 Weight. The weight of the switching unit shall be not greater than 3.5 pounds.

3.4.7 Color. The exterior color of the switching unit shall be lusterless gray. Painted surfaces shall be in accordance with FED-STD-595, Color No. 36231 except mounting surfaces shall have chemical conversion coating in accordance with MIL-C-5541.

3.4.8 Nameplates and identification marking. Nameplate approval and identification marking shall be in accordance with MIL-N-18307.

3.4.8.1. Decals, nameplates, and marking locations. All decals, nameplates, and marking shall be affixed to the switching unit as shown on Figure 1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.



Dimensions are in millimeters (inches)

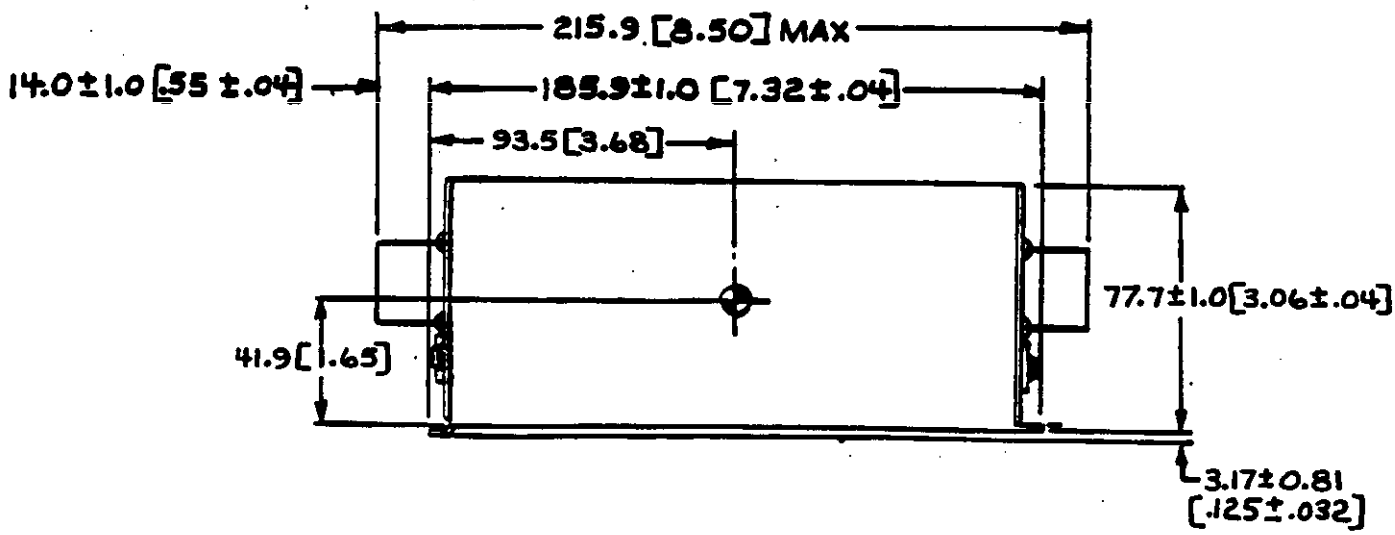


Figure 1. Outline dimensions of Switching Unit SA-2498/ARC

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

3.4.8.2 Serial numbers. Serial numbers will be assigned by the procuring activity for the switching unit under contract or purchase order. The contractor shall serialize sequentially all switching units which are delivered under the contract.

3.4.9 Input electrical power.

3.4.9.1 Operating power. The switching unit shall meet all applicable requirements of MIL-STD-704 and shall give specified performance when energized from the following power sources having characteristics and limits as specified in MIL-STD-704. The voltage supplied shall be 28 volts DC to the requirements of MIL-STD-704. The power required shall be 20 watts maximum.

3.4.9.2 Degraded performance. Degraded performance will be permitted for voltage transients as specified by MIL-STD-704 with durations of 75 microseconds to 0.5 second during normal electrical system operation. Operation shall return to normal within five seconds with no resulting damage to the switching unit.

3.4.9.3 Emergency conditions. Under emergency power conditions, the switching unit shall not be required to function in the automatic relay (plain or secure voice) modes. However, the transmit override feature of the automatic relay (plain and secure voice) mode shall be operative, whereby actuation of either radio set or receiver-transmitter external transmit (key) switch shall cause both to transmit the same signals simultaneously.

3.4.9.4 Undervoltage protection. The switching unit shall not be damaged by voltages below the minimums specified by MIL-STD-704, and shall be capable of resuming normal operation when the voltage returns within normal limits.

3.4.9.5 Reverse polarity protection. The switching unit shall be protected such that it will not be damaged by application of reverse polarity voltages.

3.4.9.6 Loss of input electrical power. Under loss of input electrical power, the switching unit shall be required to function in the plain or off mode.

3.4.10 Manufacturing requirements. The switching unit shall be manufactured in accordance with manufacturing processes, workmanship, and quality assurance provisions of MIL-STD-454, MIL-STD-275, MIL-P-55110, WS-6536, NAVMAT P-4855.1, and NAVMAT P-9492.

3.5 Performance. Unless otherwise specified herein, values set forth to establish specified performance apply to performance under both standard and extreme service and input power conditions. When reduced performance under the extreme conditions is acceptable, tolerances or values setting forth acceptable variations from the performance under the standard conditions will be specified herein.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

3.5.1 Operation. The switching unit shall provide the functions and capabilities specified in 3.1 for operation in military aircraft and helicopters. Additionally the switching unit shall be compatible with associated equipment listed in 6.5 to provide operations specified in 3.6.

3.6 Detail requirements.

3.6.1 Switching Unit, SA-2498/ARC. The switching unit shall be composed of three major components:

a. Plain text subassembly

b. Cipher text subassembly

c. Chassis subassembly

3.6.1.1 Modes of operation. The switching unit function is dependent upon the position of the mode switch on the auto-relay control unit.

3.6.1.1.1 Plain or off mode. In this mode labeled "off" on the auto-relay control, the switching unit shall provide interconnections so that both radio sets shall operate independently and shall both operate in plain voice mode.

3.6.1.1.2 Plain relay mode. The switching unit shall provide interconnection and transfer of voice and control signals for automatic relaying of plain voice. When either AIC PTT (RT1 or RT2) is activated, the relay shall be temporarily interrupted and both radios shall be keyed simultaneously. The keying operators microphone output shall be connected via the switching unit to the audio input of both radios for simultaneous modulation of the transmitters. See MIL-R-85664 and MIL-R-85665 for further description of the relay operation.

3.6.1.1.3 Crypto relay mode. The switching unit shall provide interconnection and transfer of cipher text and control signals for automatic relaying of secure voice. The radio which is actively receiving the relay "up link" signal shall transfer the demodulated signal to the switching unit. (The switching unit shall route the audio to KY-1 to decipher.) If the switching unit determines that the demodulated signal is a legitimate cipher text signal, then the switching unit routes the signal to the other R/T to transmit a relay "down link". When either microphone PTT pushbutton is activated, the switching unit temporarily interrupts the relay, and simultaneously keys both radios. The keying operators microphone output shall be directed by the switching unit to KY-1, with the resulting cipher text routed to the audio input of both radios for simultaneous modulation of the transmitters. See MIL-R-85664 and MIL-R-85665 for further description of relay operation.

3.6.1.1.4 Crypto 1 mode. The switching unit shall interconnect the equipment complement so that both radios function independently. The radio designated R/T-1 shall be connected to KY-1 by the switching unit so that it shall function in secure voice configuration. The radio designated R/T-2 shall be connected so it will function in plain voice configuration.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

3.6.1.1.5 Crypto 2 mode. The switching unit shall interconnect the equipment complement so that both radios shall function independently. The radio designated R/T-1 shall be connected so it will function in plain voice configuration. The radio designated R/T-2 shall be connected by the switching unit so the radio set will function in a secure voice configuration using KY-1 if only one TSEC/KY-28/58 is used or KY-2 if two TSEC/KY-28/58's are used.

3.6.1.1.6 Crypto 1 + 2 mode. The switching unit shall interconnect the equipment complement so that both radios will function independently. The radio designated R/T-1 shall be coupled by the switching unit to KY-1 and microphone 1 so the radio set shall function in a secure voice configuration. Similarly, the radio designed R/T-2 shall be coupled by the switching unit to KY-2 and microphone 2 so the radio set shall function in a secure voice configuration.

3.6.1.2 Signal isolation requirement. No plain text voltage in the frequency range of 300-3000 HZ shall be present on a cipher text signal channel, except to the extent that the coupled amplitude is more than 100 decibels below the peak signal found in the above referenced plain text signal channel.

3.6.1.3 HAVE QUICK compatibility. HAVE QUICK compatibility will be a design goal for KY-58 cipher relay.

3.6.1.4 Wideband audio to R/T channels. Wideband audio from R/T-1 or R/T-2 to wideband audio to R/T-2 or R/T-1 shall be a digital data channel in the cipher text module. This channel has the following characteristics:

a. The nominal gain of 30 dB.

b. Voltage amplitude limiting circuitry which limits the output amplitude to 12 volts peak to peak $\pm$ 1 volt.

3.6.1.5 Deciphered audio to R/T channels. Deciphered audio from KY-1 or KY-2 to deciphered audio to R/T-1 or R/T-2 shall an audio voice channel on the plain text module. This channel has a gain of 0 dB, -2 dB.

3.6.1.6 Microphone audio to R/T channels. Audio from microphone 1 or 2 to microphone audio to R/T-1 or R/T-2 shall be an audio voice channel on the plain text module. This channel has a gain of 0 dB, -2 dB.

3.6.1.7 Microphone audio to KY channels. Audio from microphone 1 or 2 to microphone audio to KY-1 or KY-2 is an audio voice channel on the plain text module. This channel has a gain of 0 dB, -2 dB.

3.6.1.8 Retransmit audio to R/T channels. Plain retransmit audio 1 or 2 to retransmit audio to R/T-2 or R/T-1 shall be an audio voice channel. The audio voice channel has a gain of 0 dB $\pm$ 1 dB.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the current version before use.

3.6.1.9 Bypass audio to KY channels. Plain retransmit audio 1 or 2 to bypass audio to KY-1 or KY-2 shall be an audio voice channel on the plain text module. The audio voice channel has a gain of 0 dB ± 1 dB.

3.6.1.10 External connections. Connections to external circuits shall be in accordance with Tables V and VI.

3.7 Safety. The switching unit safety requirements shall be in accordance with MIL-E-5400 except resistance from ground to chassis shall be not greater than 20 milliohms.

3.7.1 Human performance/human engineering. MIL-STD-1472 and MIL-H-46855 shall be used as a design guide for the switching unit and all component parts therein.

3.8 Bonding. Bonding shall be in accordance with MIL-B-5087.

3.9 Workmanship. Workmanship for the switching unit shall conform with MIL-STD-454, Requirement 9.

Table V. External Connections

Reference Description	Receptacle Type	Reference
J1	MIL-C-38999 Series III Special Filtered Connector Bendix P/N 21-529017-200	Cipher Text Input/Out Signals
J2	MIL-C-38999 Series III Special Filtered Connector Bendix P/N 21-529017-202	Plain Text Input/Out Signals

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the current version before use.

Table VI. Switching Unit J1 Interface Characteristics

Connector Pin No.	Function	Input-I Output-O	Characteristics
J1-1	Plain REXMT Audio to Radio 1	O	Plain Voice - 300-3500 Hz 6.5 VRMS Typical
J1-2	Spare		
J1-3	MIC 1 PTT	I	Ground When Operator Activated Push-To-Transmit Switch.
J1-4	Bypass Audio to KY-2	O	Plain Audio 300-3500 Hz 6.5 VRMS Typical
J1-5	WB XMT Audio to Radio 1	O	BB or DP Cipher Text 12V P-P Typical
J1-6	REXMT Audio from Radio 2	I	Plain Voice - 300-3500 Hz 6.5 VRMS Typical
J1-7	BB/DP Sel for KY-1 (To SA)	I	Ground for DP Mode and Open Circuit for BB Mode
J1-8	+28V Crypto 2	I	+28VDC Switched from C-11628 when either Crypto 2 or Crypto 1 + 2 Mode is selected
J1-9	RCV Cipher Text to KY-1	O	BB or DP Cipher Text 8.3V RMS Max
J1-10	WB XMT Audio to Radio 2	O	BB or DP Cipher Text 12V P-P Typical
J1-11	+28V Crypto 1	I	+28 VDC Switched from C-11628 in Crypto 1, and Crypto 1 + 2 Modes
J1-12	MIC 2 PTT	I	Ground when Operator Activates Push-to-Transmit Switch
J1-13	REXMT Control from Radio I	I	Squelch Ground from Radio Set when receiving Plain Voice Signal

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the current version before use.

Table VI. Switching Unit J1 Interface Characteristics (Continued)

Connector Pin No.	Function	Input-I Output-O	Characteristics
J1-14	Spare		
J1-15	Plain REXMT Audio to Radio 2	O	Plain Voice - 300-3500 Hz 6.5 VRMS Typical
J1-16	Spare		
J1-17	RCV Cipher Text to KY-2	O	BB or DP Cipher Text 8.3V RMS Max
J1-18	Bypass Audio to KY 1	O	Plain Audio 300-3500 Hz 6.5 VRMS Typical
J1-19	Spare		
J1-20	KY-28/58 SEL	N/A	System Configuration Jumper. To be jumpered to J1-21 only if KY's are KY-28
J1-21	Signal Ground	N/A	To be jumpered to J1-20 only if KY-28's are in system
J1-22	Spare		
J1-23	REXMT Control from Radio 2	I	Squelch Ground From Radio Set when receiving Plain Voice Signal
J1-24	REXMT Audio from Radio 1	I	Plain Voice - 300-3500 Hz 6.5 VRMS Typical
J1-25	Power Ground		
J1-26	Spare		
J1-27	KY-28 1 BB Cipher Text	I	BB Cipher Text from KY-28 12V P-P Typical
J1-28	KY-28 1 DP Cipher Text	I	DP Cipher Text from KY-28 12V P-P Typical
J1-29	KY-28 2 BB Cipher Text	I	BB Cipher Text from KY-28 12V P-P Typical

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

Table VI. Switching Unit J1 Interface Characteristics (Continued)

Connector Pin No.	Function	Input-I Output-O	Characteristics
J1-30	KY-28 2 DP Cipher Text	I	DP Cipher Text 12V from KY-28 P-P Typical
J1-31	PTT to Radio 1	O	Ground when Radio 1 to be in Transmit Mode
J1-32	Open (Reserved)		
J1-33	Spare		
J1-34	KY-28 2 BB/DP Select	I	Ground when KY-28 used and DP Mode selected for KY-2
J1-35	KY-28 1 BB/DP Select	I	Ground when KY-28 used and DP Mode selected for KY-1
J1-36	System Configuration Jumper Interconnect	N/A	To be jumpered to J1-37 when System only has 1 KY
J1-37	Signal Ground	N/A	To be jumpered to J1-36 when System only has 1 KY
J1-38	Spare		
J1-39	PTT from KY-58 2	I	Ground when KY-58 is in XMT Mode
J1-40	+28V Plain Relay	I	+28VDC switched from C-11628 when Plain Relay Mode is Selected
J1-41	+28V Crypto Relay	I	+28VDC switched from C-11628 when Crypto Relay Mode is Selected
J1-42	Spare		
J1-43	PTT from KY-58 1	I	Ground when KY-58 is in XMT Mode
J1-44	PTT to Radio 2	O	Ground when Radio 2 to be in XMT Mode

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

Table VI. Switching Unit J1 Interface Characteristics (Continued)

Connector Pin No.	Function	Input-I Output-O	Characteristics
J1-45	PTT to KY-1	O	Ground when KY-1 to be in XMT Mode
J1-46	Spare		
J1-47	Cipher text from KY-58 1	I	BB or DP Cipher Text from KY-58. 12V P-P Typical
J1-48	Reserved		
J1-49	PTT to KY-2	O	Ground when KY-2 to be in XMT Mode
J1-50	Spare		
J1-51	Cipher text from KY-58 2	I	BB or DP Cipher Text from KY-58, 12V P-P Typical
J1-52	Spare		
J1-53	WB RCV Audio from Radio 1	I	BB or DP Cipher Text 8.3 RMS Max
J1-54	WB RCV Audio from Radio 2	I	BB or DP Cipher Text 8.3V RMS Max
J1-55	Shield Ground	N/A	

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:51Z
Check the source to verify that this is the current version before use.

Table VII. Switching Unit J2 Interface Characteristics

Connector Pin No.	Function	Input-I Output-O	Characteristics
J2-1	Spare		
J2-2	Spare		
J2-3	MIC Audio to Radio 2 (High)	O	Plain Voice 300-3500 Hz 0.35-0.5 VRMS Typical High Side of Balanced Output
J2-4	Signal Ground	N/A	To be jumpered to (J2-19) X-Mode RCV Audio to Radio 1 Low Side for System with Unbalanced Input/Output
J2-5	X-Mode RCV Audio to Radio 1 (High)	O	Plain Voice 300-3500 Hz 6.5 VRMS Typical. High side of Balanced Output
J2-6	MIC Audio to Radio 2 (Low)	O	Low Side of Balanced MIC Audio 2 Output. Jumper to Pin J2-7 for Unbalanced Systems
J2-7	Signal Ground	N/A	To be jumpered to (J2-6) MIC Audio to Radio 2 Low Side for System with Unbalanced Output
J2-8	MIC Audio to KY-1 (High)	O	Plain Voice 300-3500 Hz 0.35-0.5 VRMS. Typical High Side of Balanced Output
J2-9	MIC Audio to KY-2 (High)	O	Plain Voice 300-3500 Hz 0.35-0.5 VRMS Typical High Side of Balanced Output
J2-10	Deciphered Audio from KY-2 (High)	I	Plain Voice 300-3500 Hz 6.5 VRMS Typical. High Side of Balanced Input
J2-11	MIC 1 PTT	I	Ground when Operator Activates Push-to-Transmit Switch
J2-12	Signal Ground	N/A	To be jumpered to (J2-14) MIC Audio to KY-1 Low Side for System with Unbalanced Output

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the current version before use.

Table VII. Switching Unit J2 Interface Characteristics (Continued)

Connector Pin No.	Function	Input-I Output-O	Characteristics
J2-13	MIC Audio to Radio 1 (High)	O	Plain Voice 300-3500 Hz 0.35-0.5 VRMS Typical. High Side of Balanced Output.
J2-14	MIC Audio to KY-1 (Low)	O	Low Side of Balanced MIC Audio 1 Output. Jumper to Pin J2-12 for Unbalanced Systems
J2-15	MIC 2 PTT	I	Ground when Operator Activates Push-to-Transmit Switch
J2-16	Signal Ground	N/A	To be Jumpered to (J2-24) MIC Audio to KY-2 Low Side for System with Unbalanced Output
J2-17	System Configuration Jumper Interconnect	N/A	To be Jumpered to (J2-10) Deciphered Audio from KY-2 for system with only 1 KY
J2-18	X-Mode RCV Audio to Radio 2 (High)	O	Plain Voice 300-3500 Hz 6.5 VRMS Typical. High Side of Balanced Output
J2-19	X-Mode RCV Audio to Radio 1 (Low)	O	Plain Voice 300-3500 Hz 6.5 VRMS Typical. Low Side of Balanced Output. Jumper to Pin J2-4 for Unbalanced Systems
J2-20	MIC Audio to Radio I (Low)	O	Low Side of Balanced MIC Audio 1 Output. Jumper to Pin J2-21 for Unbalanced Systems
J2-21	Signal Ground	N/A	To be Jumpered to (J2-20) MIC Audio to Radio 1 Low Side for System with Unbalanced Output
J2-22	Spare		
J2-23	PTT from KY-58 1	I	Ground when KY-58 is in XMT Mode

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:51Z
Check the source to verify that this is the current version before use.

Table VII. Switching Unit J2 Interface Characteristics (Continued)

Connector Pin No.	Function	Input-I Output-O	Characteristics
J2-24	MIC Audio to KY-2 (Low)	O	Low Side of Balanced MIC Audio 2 Output. Jumper to Pin J2-16 for Unbalanced Systems
J2-25	Power Ground	N/A	
J2-26	Deciphered Audio from KY-2 (Low)	I	Plain Voice 300-3500 Hz 6.5 VRMS Typical. Low Side of Balanced Input
J2-27	X-Mode RCV (Low) Audio to Radio 2	O	Plain Voice 300-3500 Hz 6.5 VRMS Typical. Low Side of Balanced Output. Jumper to Pin J2-34 for Unbalanced Systems
J2-28	System Configuration Jumper Interconnect	N/A	To be Jumpered to Pin J2-36 for System with Only 1 KY
J2-29	Spare		
J2-30	Spare		
J2-31	Spare		
J2-32	Open (Reserved)		
J2-33	System Configuration Jumper Interconnect	N/A	To be Jumpered to (J2-26) Deciphered Audio from KY-2 for System with Only 1 KY
J2-34	Signal Ground	N/A	To be Jumpered to (J2-27) X-Mode RCV Audio to Radio 2 Low Side for System with Unbalanced Input/Output
J2-35	Spare		
J2-36	Signal Ground	N/A	To be Jumpered to Pin J2-28 for System with Only 1 KY

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the current version before use.

Table VII. Switching Unit J2 Interface Characteristics (Continued)

Connector Pin No.	Function	Input-I Output-O	Characteristics
J2-37	Spare		
J2-38	Spare		
J2-39	Spare		
J2-40	Deciphered Audio from KY-1 (Low)	I	Plain Voice 300-3500 Hz 6.5 VRMS Typical. Low Side of Balanced Input
J2-41	Deciphered Audio from KY-1 (High)	I	Plain Voice 300-3500 Hz 6.5 VRMS Typical. High Side of Balanced Input
J2-42	MIC Audio 1 (Low)	I	Low Side of Balanced MIC Audio 1 Input. Jumper to Pin J2-44 for Unbalanced Systems
J2-43	Spare		
J2-44	Signal Ground	N/A	To be Jumpered to (J2-42) MIC Audio 1 Low Side for Systems with Unbalanced Input
J2-45	System Configuration	N/A	To be Jumpered to (J2-46) MIC Audio 2 Low Side for System with Unbalanced Input
J2-46	MIC Audio 2 (Low)	I	Low Side of Balanced MIC Audio 2 Input. Jumper to Pin J2-45 for Unbalanced Systems
J2-47	Spare		
J2-48	Spare		
J2-49	MIC Audio 1 (High)	I	Balanced Input Plain Voice Audio 300-3500 Hz 0.35-0.5V RMS Typical

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:51Z
Check the source to verify that this is the currentversion before use.

Table VII. Switching Unit J2 Interface Characteristics (Continued)

Connector Pin No.	Function	Input-I Output-O	Characteristics
J2-50	+28V Crypto 1	I	28VDC Switched from C-11628 in Crypto 1, and Crypto 1 + 2 Modes
J2-51	+28V Plain Relay	I	+28VDC Switched from Control C-11628 when Plain Relay Mode is Selected
J2-52	MIC Audio 2 (High)	I	Balanced Input Plain Voice Audio 300-3500 Hz 0.35-0.5 VRMS Typical
J2-53	+28V Crypto Relay	I	+28VDC Switched from Control C-11628 when Crypto Relay Mode is Selected
J2-54	+28V Crypto 2	I	+28VDC Switched from C-11628 when either Crypto 2 or Crypto 1 + 2 Mode is Selected
J2-55	Shield Ground	N/A	

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Standard conditions. Unless otherwise specified herein, all inspections shall be performed with conditions specified as follows:

a.	Temperature	Room ambient (25 $\pm$ 10°C)
b.	Altitude	Normal ground
c.	Vibration	None
d.	Humidity	Room ambient up to 90 percent relative humidity
e.	Input Power	28 $\pm$ 0.5 volt dc

4.4 First article inspection. First article inspection shall be conducted by the contractor on switching unit representative of the production switching unit to be supplied under the contract. First article tests shall be accomplished under the authorized test procedures (see 4.6). No first article inspections shall be conducted prior to acceptance of the first article test procedure by the procuring activity. First article inspection shall consist of all tests described in 4.4.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

4.4.1 Scope of first article tests. First article tests shall include all tests in the approved test procedures to determine that the switching unit meets all the requirements of this specification and of the contract. First article tests shall include environmental tests in accordance with the procedures of MIL-T-5422 including temperature shock, a random vibration test as specified in 4.4.4, and an electromagnetic compatibility test in accordance with MIL-STD-461 and MIL-STD-462, as modified in 4.4.3. The first article test may also include a reliability qualification test.

4.4.2 First article approval. Approval of the first article sample will be by the procuring activity upon completion of all tests.

4.4.3 Electromagnetic interference tests. Compliance with the requirements of 3.4.4.3 shall be demonstrated by tests. These tests shall be performed in accordance with MIL-STD-462, with Notice 4 as modified herein.

4.4.3.1 EMI test plan. The contractor shall perform EMI tests in accordance with a contractor prepared EMI Test Plan. The Test Plan shall describe all EMI tests required to demonstrate compliance with the requirements of 3.4.4.3, and shall address all anomalies, inconsistencies, and omissions (if any). The Test Plan shall detail operating modes to be tested. The Test Plan shall be acceptable to the procuring activity prior to the commencement of EMI testing.

4.4.3.2 Ground plane interference. The contractor shall prepare a procedure to demonstrate that the switching unit complies with the requirements of 3.4.4.3.4. The procedure shall be included in the EMI Test Plan.

4.4.3.3 Conducted susceptibility test, interconnecting and signal leads. In addition to the requirements of MIL-STD-461 the following RF susceptibility test shall be conducted. No malfunction, undesirable response, or change in indication beyond the tolerance given in the equipment specification or approval test plan shall be produced in any equipment when the RF signal levels listed in Table VII are coupled into the interconnecting wiring.

Table VIII. Susceptibility test signals

Frequency	Required Signal Levels In One Turn Loop
2 to 30 MHz	100 MV RMS

The RF signal shall be modulated 80 percent with 400 Hz and with that frequency and type of modulation to which the test sample is most susceptible. This requirement applies to all interconnecting wires or cable between component parts of subsystems, and to wires and cables connected to other equipment. The RF signal shall be measured in the one-turn loop with an RF voltmeter as shown on Figure 2. The signal generator shall have an output

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the current version before use.

impedance of 50 ohms or less and a maximum open-circuit voltage of not less than 3 volts rms. The test procedure shall be as follows:

a. With the current probe clamped around the wire bundles under test, set the signal generator for maximum output and slowly scan from 2 MHz to 30 MHz.

b. At those frequencies where a response is noted, move the current probe along the wire for maximum indication. Reduce the signal generator output and determine the threshold of susceptibility of the wire bundle. When a response is found on an interconnecting wire bundle, probe each wire in the bundle separately to determine the susceptible wire(s).

c. Record and include in the test report the threshold of susceptibility data (frequency, amplitude, and description of responses) obtained in (b) above. Also include in the test report the corrective action required for each wire that does not meet the required level.

4.4.3.3.1 Wire bundles. Wire bundles may be tested in sections if the current probe will not fit around the complete wire bundles. Chassis or circuit grounds not normally routed with the interconnecting wiring shall be measured individually and shall not be included in the wire bundle under test. Power ground return leads shall not be placed in the probe.

4.4.3.4 CE03 test procedure. The CE03 requirement of 3.4.4.3.1 shall be demonstrated using MIL-STD-462, Notice 2, Test Method CE03 for all power leads and MIL-STD-462, Notice 2, Test Method CE04 for all control and signal leads tested.

4.4.3.5 CE07 test procedure. The contractor shall prepare a test to demonstrate compliance with the CE07 requirement of 3.4.4.3.1. This test shall be included as a part of the EMI Test Plan.

4.4.4 Random vibration test. A random vibration test shall be conducted as specified in 4.4.4.1 through 4.4.4.4.

4.4.4.1 Test item operation. The test item shall be operated during application of random vibration so that the functional effects caused by these tests may be evaluated. The test item shall meet performance requirements, as specified, while the functional vibration levels are being applied and immediately preceding and following the application of the endurance levels.

4.4.4.2 Mounting techniques. The test item shall be hard mounted either directly to the vibration exciter or transition table, or by means of a rigid fixture capable of transmitting the vibration conditions specified herein. Precautions shall be taken in the establishment of mechanical interfaces to minimize the introduction of extraneous responses in the test setup. The test load shall be distributed as uniformly as possible on the vibration excitertable in order to minimize effects of unbalanced loads. The input control sensing device(s) shall be rigidly attached to the vibration table, or fixture if used, as near as possible to the attachment point(s) of the test item. Additional vibration sensors shall be located in or on the test item to determine resonant frequencies and amplification factors. Locations to be

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the currentversion before use.

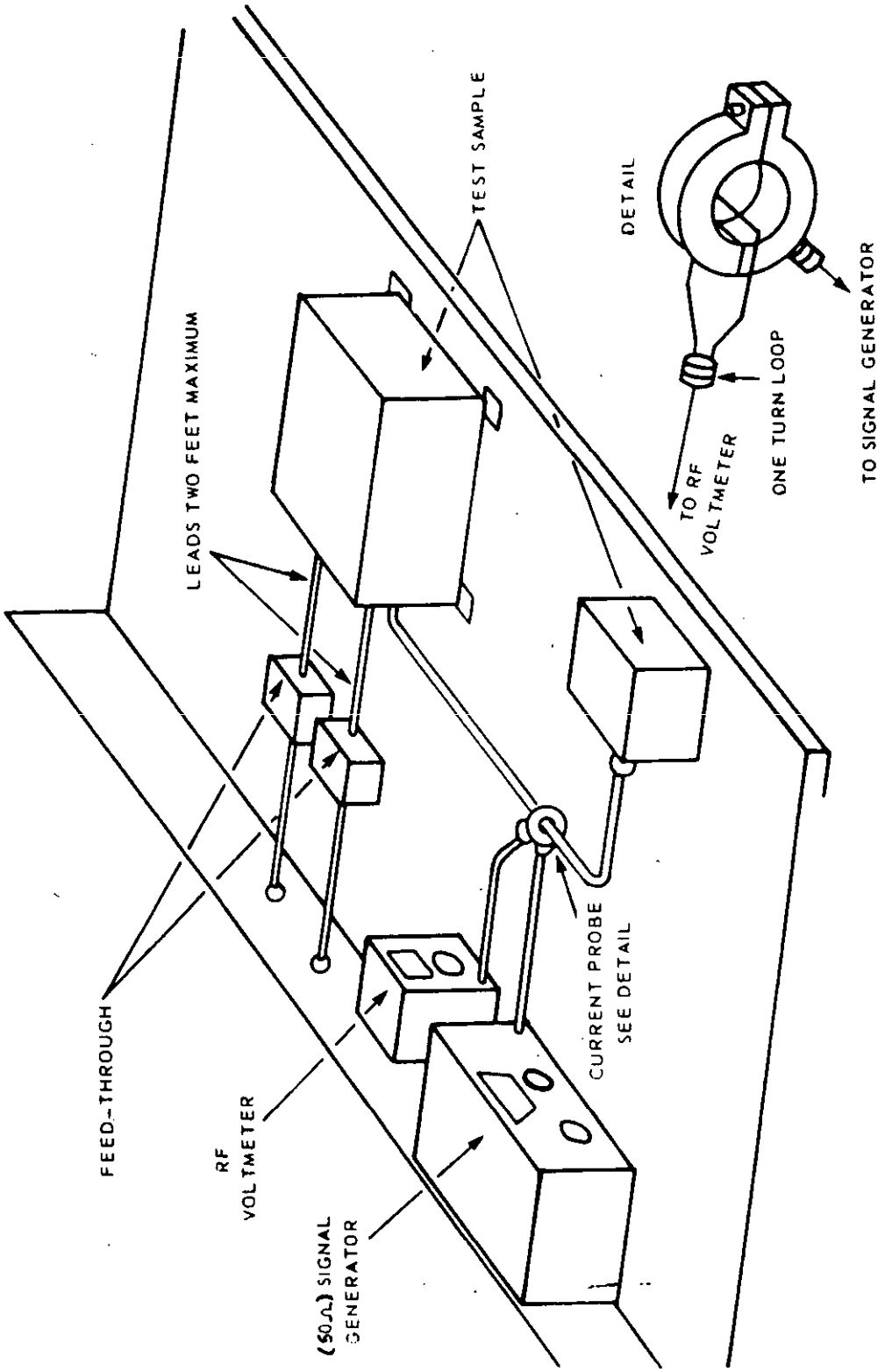


FIGURE 2. Conducted susceptibility test set up

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the current version before use.

selected should include main structure, printed circuit boards, large components, and modules, where practicable. The sensor sizes and weights shall be limited so that their effect on the dynamic responses being measured is minimal.

4.4.4.3 Performance of test. The individual equipment test item shall be subjected to broadband random vibration excitation with the power spectral density envelope shown on Figure 3. The test item shall be attached to the vibration exciter according to 4.4.4.2. Vibration shall be applied sequentially along each of the three orthogonal axes of the test item. Two test levels are required, a functional level and an endurance level. For each axis, one half of the functional test shall be conducted first, then the endurance test, followed by the second half of the functional test. The equipment shall perform according to the operating requirements as specified in 4.4.4.1. The acceleration power spectral density (g^2/Hz) of applied vibration, as measured on the test fixture at mounting points of the test item, shall be as specified in 4.4.4.4. Test times shall, for each axis, be

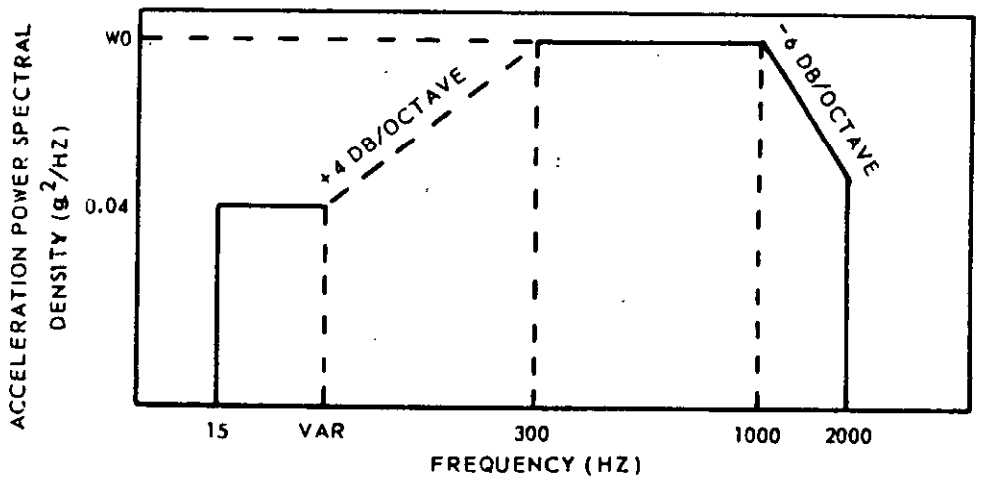


Figure 3. Random vibration envelope

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the current version before use.

one hour each for functional and endurance levels. The instantaneous random vibration acceleration peaks may be limited to three times the rms acceleration level. The power spectral density of the test control signal shall not deviate from the specified requirements by more than +100, -30 percent (+3, -1.5 dB) below 500 Hz and +100, -50 percent (+3 dB) between 500 Hz and 2,000 Hz except that deviations as large as +300, -75 percent (+6 dB) shall be allowed over a cumulative bandwidth of 100 Hz maximum, between 500 and 2,000 Hz. Tolerance levels in terms of dB are specified as:

$$dB = 10 \text{ Log}_{10} \frac{W1}{W0}$$

Where W1 = measured acceleration power spectral density in g^2/Hz units, the term W0 defines the specified level in g^2/Hz units. Confirmation of these tolerances shall be made by use of an analysis system providing statistical accuracies corresponding to a bandwidth-time constant product, BT=50 minimum. Specific analyzer characteristics shall be as specified below or equivalent, subject to the BT=50, minimum limitation.

a. On-line, contiguous filter, equalization analysis system having a bandwidth as follows:

B=25 Hz, maximum between 20 and 200 Hz
B=50 Hz, maximum between 200 and 1,000 Hz
B=100 Hz, maximum between 1,000 and 2,000 Hz

b. Swept frequency analysis systems characterized as follows:

(1) Constant bandwidth analyzer.

(a) Filter bandwidth as follows:

B=25 Hz, maximum between 20 to 200 Hz
B=50 Hz, maximum between 200 to 1,000 Hz
B=100 Hz, maximum between 1,000 to 2,000 Hz

(b) Analyzer averaging time =T= 2 RC = 1 second, where T= true averaging time constant

(c) Analysis sweep rate (linear) =

$$R = \frac{B}{4RC} \text{ or } \frac{B^2}{8} \quad \text{(Hz/second)}$$

(2) Constant percentage bandwidth analyzer

(a) Filter bandwidth - pfc=one-tenth of center frequency maximum (0.1 fc) where p=percentage and fc=analyzer center frequency

(b) Analyzer average time = T = $\frac{50}{pfc}$, minimum

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the current version before use.

(c) Analysis sweep rate (logarithmic) =

$$T = \frac{pfc}{4RC} \text{ or } \frac{(pfc)^2}{8}$$

(Hz/second), maximum, whichever is smaller

Digital power spectral density analysis employing quantization techniques providing accuracies corresponding to the above approach.

4.4.4.3.1 Worst case axis. During the vibration testing, data shall be obtained to identify the worst case axis for purpose of vibration screening tests on production equipments.

4.4.4.4 Test levels. For the purpose of the random vibration test, the following test levels shall be used:

a. Functional Level-W_o = 0.05 g^2/Hz (8.5g rms minimum)

b. Endurance Level-W_o = 0.12 g^2/Hz (12.7 g rms minimum)

4.4.5 Reliability qualification test. A reliability qualification test may be included in the first article tests. A minimum of six switching unit shall be tested in accordance with MIL-STD-781 using the reliability qualification test - mission profile in this specification. (See Figures 4 and 5.) Each switching unit shall have successfully completed the normal burn-in and all equipment test (AET) prior to the start of RQT. The mission profile test cycle should begin at the "12 hour" point which is a cold day ground stabilization (-54°C) prior to ground operation and flight in a cold day. Random vibration should be applied with the units hard-mounted (without vibration isolators). The accept-reject criteria imposed shall be those of MIL-STD-781, Test Plan IIIC, with the upper test MTBF specified in 3.4.1.3, except that the test shall be truncated after 10,300 hours.

4.4.6 Production switching unit. Switching units supplied under the contract shall in all respects, including design, construction, workmanship, performance and quality, be capable of successfully passing the same tests as specified herein.

4.5 Quality conformance inspection. The contractor shall furnish all samples and shall be responsible for accomplishing the quality conformance tests. All inspection and testing may be under the supervision of the customer representative. The contractor shall retain test data showing quantitative results for all quality conformance tests. Such tests shall be signed and stamped by an authorized representative of the contractor or laboratory, as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Quality conformance tests shall consist of the following:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

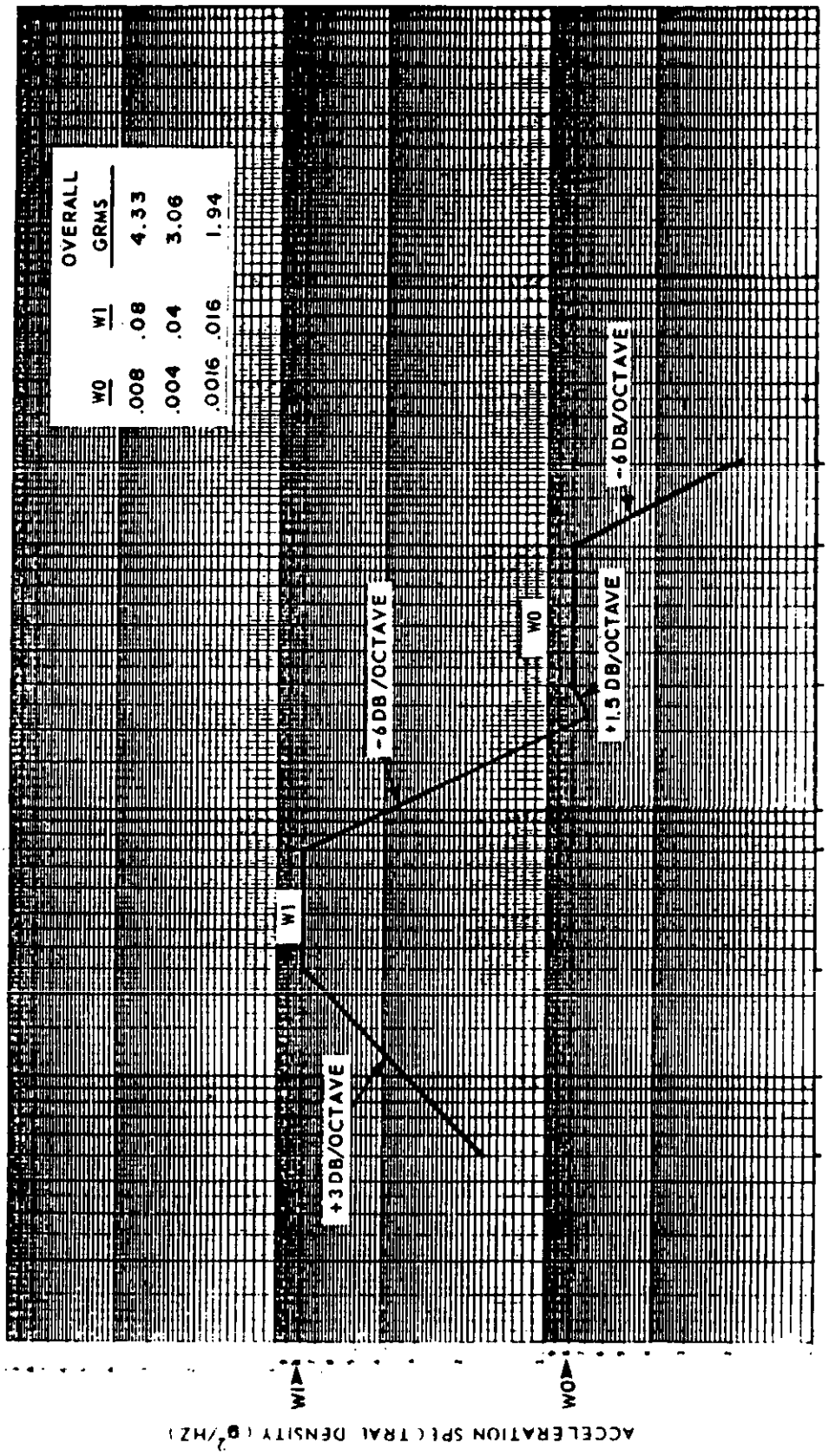


FIGURE 4. RQT vibration envelope (see 4.4.5)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

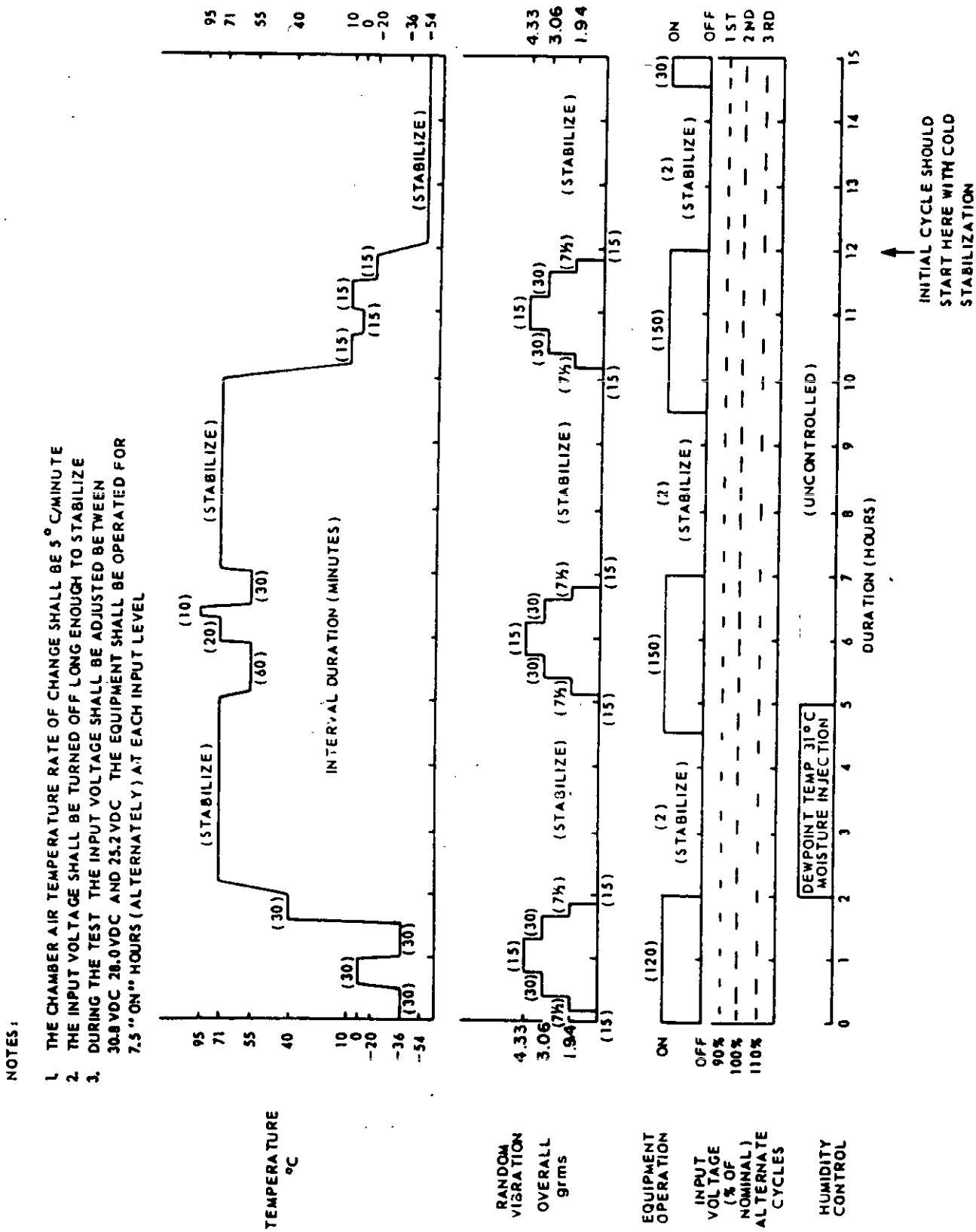


Figure 5. Temperature/power cycle (see 4.4.5).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

a. Individual tests (see 4.5.1)

b. Manufacturing screening (see 4.5.2)

c. Sampling tests (see 4.5.3)

d. Reliability assurance tests (see 4.5.4)

e. Special tests (see 4.5.5)

4.5.1 Individual tests. Each switching unit submitted for acceptance shall be subjected to the individual tests. The tests shall be adequate to determine compliance with the requirements of materials, workmanship, operational adequacy and reliability. As a minimum, each switching unit accepted shall have passed the following tests:

a. Examination of product (see 4.5.1.1)

b. Operational test (see 4.5.1.2)

4.5.1.1 Examination of product. Each switching unit shall be examined carefully to determine that the material and workmanship requirements have been met.

4.5.1.2 Operational test. Each switching unit shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record data to assure specified equipment operation.

4.5.2 Manufacturing screening. The contractor shall subject each WRA to manufacturing screening and burn-in prior to undergoing sample tests or reliability assurance tests. Manufacturing screening shall consist of WRA random vibration and temperature cycling. An additional screening period may be used at the option of the contractor. If an additional screening period is to be used, the details thereof shall be included in the approved test procedures and shall be the same for all equipment, including spare units.

4.5.2.1 WRA random vibration. Prior to conducting WRA temperature cycling, each WRA shall have completed a 10-minute failure-free vibration screen at 6.1g rms. Power spectral density shall cover 20-2000 Hertz. Pseudo-random vibration methods are acceptable. The vibration envelope is shown on Figure 6. The test item shall be hard mounted without vibration isolators and shall be operating throughout the test.

4.5.2.2 WRA temperature cycling. Each WRA shall undergo 25 temperature cycles between -54°C and 71°C. The rate of temperature change between extremes shall be at least 5°C per minute. When the WRA temperature has stabilized within 5°C or either temperature extreme, the WRA shall be soaked. Hot soak at +71°C shall be two hours; cold soak at -54°C duration shall be one hour. At the conclusion of each soak, the next temperature ramp shall begin.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

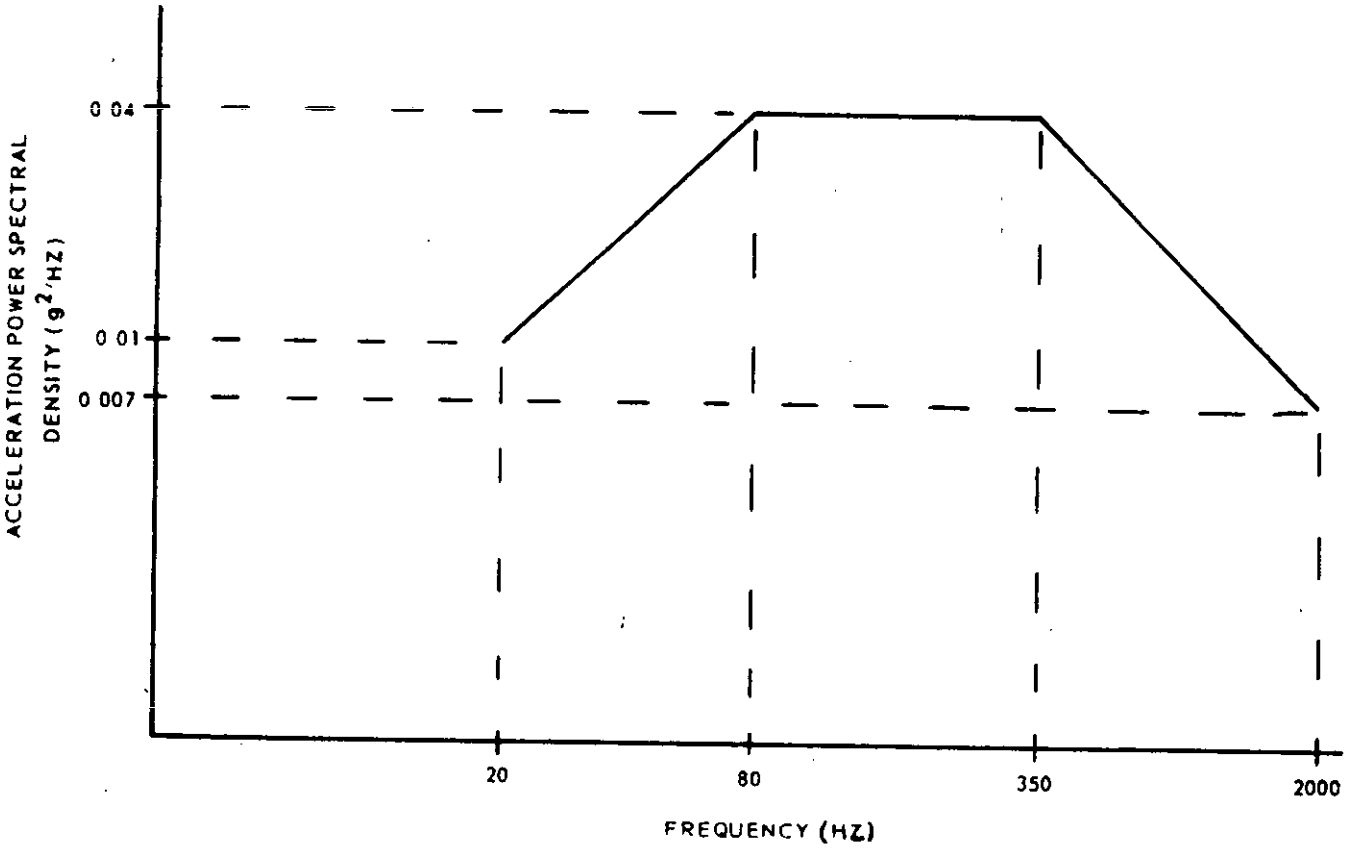


Figure 6. Manufacturing screening random vibration envelope

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

The WRA shall be energized throughout temperature cycling except during the cooldown period from +71°C to -54°C and during the 60 minute soak at -54°C. In addition, the WRA shall be turned off for one minute at points 30, 60, and 90 minutes into each hot soak period. Switching unit performance shall be periodically checked during each cycle and repairs shall be made to correct failures. The last five consecutive temperature cycles shall be failure free. One temperature cycle is defined as: operation at +71°C for the 120 minute hot soak (including three minutes "off"), non-operational cooldown to -54°C, 60 minute non-operational cold soak at -54°C, operational temperature increase to +71°C.

4.5.3 Sampling tests. Switching units shall be subjected to sampling tests. Switching units selected for sampling tests shall first have passed the individual tests. Switching units will be selected for sampling tests by the government inspector in accordance with Table IX.

Table IX. Sampling test sample selection

Quantity of equipments offered for acceptance	Quantity to be selected for sampling test
First 10	1
Next 50	1
Next 75	1
Next 100	1
	1 for each additional 200 or fraction thereof

Sampling tests are not required when reliability assurance tests are conducted.

4.5.3.1 Scope of sampling tests. As a minimum, each switching unit selected for sampling tests shall be subjected to the following tests:

a. Complete operational test at ambient room conditions, making all necessary measurements to assure that all applicable specification requirements have been met.

b. Operational test at certain environmental conditions. The conditions may vary for each switching unit tested and should be based on results of the first article, individual and special tests.

c. During the reliability sample test specified in 4.5.4.2, each failure shall be analyzed as to cause and remedial action necessary to reduce the possibility of its recurring in future equipment taken.

4.5.4 Reliability assurance tests. A reliability assurance test shall be conducted using MIL-STD-781. Classification of failure shall be in accordance with 4.5.7.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the current version before use.

4.5.4.1 All equipment test. Each switching unit shall be tested as specified in MIL-STD-781 and modified herein. Temperature cycling and power on/off cycling shall as shown on Figure 5. No vibration is required. Each switching unit shall have successfully completed manufacturing screening as specified in 4.5.2 prior to the start of the test. Each equipment (WRA) shall be tested for 125 equipment operating hours with the last two cycles failure free. For determining MTBF compliance each equipment (WRA) shall be tested separately as opposed to testing as an equipment system. The test MTBF shall be as listed in 3.4.1.3. To determine whether the MTBF is being met at any time during the contract the operating time and the failures therein (not counting any burn-in failures or burn-in operating time) shall be totaled and the results compared with the reject line of Test Plan XVIIIC of MIL-STD-781. (Extend the line as necessary to accommodate the data.) These totals shall accumulate so that at any time the experience from the beginning of the contract is included. At any time that the current totals of test hours and test failures plotted on Test Plan XVIIIC curves show a reject situation, the procuring activity shall be notified. The procuring activity reserves the right to stop the acceptance of switching units at any time that a reject situation exists pending a review of the contractors efforts to improve the equipment, the equipment parts, or the equipment workmanship, so that the entire compilation will show other than a reject decision.

4.5.4.2 Reliability sample test. When specified in the contract (see 6.2.1), a monthly sample test of equipments shall replace the reliability test of 4.5.4.1. Units shall be tested as specified in MIL-STD-781 as modified herein. Temperature cycling and power on/off cycling are required and shall be as shown on Figure 5. Each sample equipment (WRA) shall be tested for 256 hours with the last two cycles failure free. For determining MTBF compliance each designated equipment (WRA) shall be tested separately as opposed to testing as an equipment system. The MTBF for each designated equipment shall be those listed in 3.4.1.3. To determine whether the MTBF is being met at any time during the contract the operating test hours and the failures therein (not counting any burn-in failures or burn-in operating time) shall be totaled and the results compared with the reject line of Test Plan IIIC of MIL-STD-781. (Extend the line as necessary to accommodate the data.) These totals shall accumulate so that at any one time the experience from the beginning of the contract is included. The sample size shall be based on the production quantity of that equipment for the month as specified in Table X.

Table X. Reliability test sample size

Production Quantity	Sample Size
2 - 5	1
6 - 25	2
26 - 50	3
51 - 100	4

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

4.5.4.3 Test details. Test details such as the performance criteria to be measured, special failure criteria, and preventive maintenance to be allowed during the test shall be part of the test procedures to be prepared by the contractor prior to the beginning of the reliability assurance tests.

4.5.5 Special tests. When specified in the contract, special tests shall be conducted for the purpose of checking the effect of any design or material change on the performance of the switching unit and to assure quality control (see 6.2.1). The switching unit selected for special tests may be selected from switching unit previously subjected to the sampling or reliability tests.

4.5.5.1 Special test schedule. Selection of switching units for special tests shall be made as follows:

a. On an early equipment after an engineering or material change.

b. Whenever failure reports or other information indicate that additional tests are required. (This will be determined by the procuring activity.)

4.5.5.2 Scope of special tests. Special tests shall consist of such tests as authorized by the procuring activity. Test procedures previously approved for the first article tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure for the procuring activity acceptance prior to conducting the tests.

4.5.6 Equipment failure. Should a failure occur during either the sampling, reliability qualification, reliability assurance or special tests, the following action shall be taken:

a. Determine the cause of failure.

b. Determine if the failure is an isolated case or design defect or pattern failure.

c. Prepare proposed corrective action intended to reduce the possibility of the same component occurring in future tests or documented justification where no corrective action is proposed.

d. Where practical, include a test in the individual test to check all switching units for this requirement until assurance is obtained that the defect has been corrected.

4.5.7 Failure classification. All incidents occurring during reliability assurance tests shall be classified and reported as either relevant or non-relevant in accordance with MIL-STD-2074 and MIL-STD-781. Only those incidents classified as relevant failures shall be accountable for making an accept/reject decision.

4.5.7.1 Relevant failures. All test incidents shall be considered as relevant failures unless determined to be non-relevant in accordance with the criteria of MIL-STD-2074.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the current version before use.

4.5.7.2 Non-relevant failures. Although non-relevant failures are not used for reliability assurance test accept or reject decisions, all test article incidents and equipment failures shall be recorded and reported.

4.6 Test procedures. The procedures used for conducting all tests required by Section 4 shall be prepared by the contractor for review and acceptance. The right is reserved by the procuring activity or the government representative to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of the specification or the contract (see 6.2.1). MIL-T-18303 shall be used as a guide for preparation of test procedures.

4.7 Testing of spares and repair parts. Equipment delivered as spare or repair parts shall undergo acceptance tests to ensure that the items meet performance and reliability requirements not less than those imposed in production.

4.7.1 Weapons replaceable assemblies. All items to be delivered as spares at the weapons replaceable assembly (WRA) or higher level shall receive performance, manufacturing screening and reliability assurance tests commensurate with that afforded the original production equipment.

4.7.2 Shop replaceable assemblies. All items to be delivered as spares at the shop replaceable assembly (SRA) or sub-SRA level shall receive functional checks and manufacturing screening under conditions and for a duration commensurate with that imposed on that assembly in the original production.

4.7.3 Repair parts. All electronic component parts to be delivered as repair parts shall receive functional and screening tests equal to those imposed for the nearest military standard part.

4.7.4 Test details. The details of acceptance and screening tests to be performed on SRA's and WRA's supplied as spare or repair parts shall be incorporated into item specifications and drawings and test procedures.

4.8 Presubmission testing. No item, part, or complete equipment shall be submitted until it has been previously tested by the manufacturer and found to comply with all applicable requirements of this specification and the contract.

5. PACKAGING

5.1 General. All major units and parts of the switching-unit shall be preserved, packaged, packed, and marked in accordance with MIL-E-17555 for the level of shipment specified in the contract or purchase order (see 6.2.1).

5.2 Marking. Shipments shall be marked in accordance with requirements of MIL-STD-129.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. The switching unit with its associated control, when operating with two radio sets and two TSEC/KY units is intended to provide an automatic relay station in plain or secure voice, simultaneous plain or secure voice transmission from each radio set, or independent plain or secure voice communication of each radio set.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification

b. If a first article inspection is required (see 3.2, 6.3)

c. If non-standard parts or materials are required (see 3.3.5)

d. If reliability program is not to be established by contractor (see 3.4.1)

e. Whether All Equipment Test or Reliability Sample Test is required (see 4.5.4.1, 4.5.4.2)

f. If Special tests are required (see 4.5.5)

g. Test procedure modifications or additional tests, if required (see 4.6)

h. Selection of applicable levels of packaging and packing (see 5.1)

6.2.2 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements identified below shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.410-6 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data specified below shall be delivered by the contractor in accordance with the contract or purchase order requirements. Deliverable data required by this specification are cited in the following paragraphs.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the current version before use.

Paragraph	Data requirement title	Applicable DID no.
3.3.2	Engineering Change Proposals (ECPs) and Requests for Deviations and Waivers (Long Form)	DI-E-2037
3.3.3	Program Parts Selection List (PPSL)	DI-MISC-80072
3.4.1, 4.5.6.c	Plan, Failure Data Collection, Analysis and Corrective Action	UDI-T-23719
3.4.1.2	Part Approval Request	DI-MISC-80071
3.4.8	Request, Approval of Identification Plate Drawings	DI-CMAN-80196
	Request for Confirmation of Nomenclature	DI-CMAN-80194
	Request for Assignment of Serial Number and Serial Number Prefix Letters	DI-CMAN-80195
	Request, Assignment of Nomenclature	UDI-E-21582
4.4, 4.6	Procedure, First Article Inspection	DI-T-4901
4.4.2	Report, First Article Inspection	DI-T-4902
4.4.3.1, 4.4.3.2, 4.4.3.5	Plan, EMI Test	DI-EMCS-80201
4.4.3.3.c	Report, EMI Test	DI-EMCS-80200
4.5.4.1, 4.5.4.2	Report, Reliability Test	DI-R-7034
4.5.4.3, 4.6	Reliability Test Procedures	DI-RELI-80251
4.5.5.2, 4.6	Procedures, Production/Acceptance Inspection	DI-T-4903
4.5.7.2	Failed Item Analysis Report	DI-RELI-80253
4.7.4	Drawings, Engineering and Associated Lists	DI-E-7031

(Data item descriptions related to this specification, and identified in section 6 will be approved and listed as such in DOD 5010.12L., AMSDL. Copies of data item descriptions required by the contractors in connection with specific acquisition functions should be obtained from the Naval Publications and Forms Center or as directed by the contracting officer.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

6.3 Options. The following options are available:

a. An additional manufacturing screening period is allowable at the option of the contractor (see 4.5.2).

b. Acceptance of equipment may be stopped anytime a reject situation exists in the All Equipment Test. This is at the option of the procuring activity (see 4.5.4.2).

6.4 First article. When a first article inspection is required, the item should be a first article sample. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Associated equipment. The switching unit should operate with the associated equipment listed in Table XI which should not be supplied as part of this equipment:

Table XI. Associated equipment

Item	Equipment Designation	Military Specification
Intercommunications Sets	AN/AIC-10, AN/AIC-14 AN/AIC-18, AN/AIC-25 C-6533/ARC	MIL-I-22353 MIL-I-27543 MIL-C-55653
Microphones, Boom Mounted	M-96A/A	MIL-M-22179
Amplifiers, Audio Frequency	AM-3597B/A	MIL-A-23595
Speech Security Equipment	TSEC/KY-28 TSEC/KY-58 TSEC/KY-58	CSEEB-13 CSEEB-32 CSEEB-14
Indicator, Frequency/ Channel	ID-2121A/ARC ID-2229A/ARC ID-2303/ARC	MIL-I-85671 MIL-I-85671 MIL-I-85671
Receiver-Transmitter, Radio	RT-1324A/ARC RT-1327A/ARC RT-1407/ARC	MIL-R-85665 MIL-R-85665 MIL-R-85665

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the current version before use.

Table XI. Associated equipment (Continued)

Item	Equipment Designation	Military Specification
Control, Switching Unit	C-11628/ARC	MIL-C-85674
	C-10777A/ARC	MIL-C-85674
	C-11132/ARC	MIL-C-85674
	C-11128/ARC	MIL-C-85806
	C-11129/ARC	MIL-C-85806
	C-11130/ARC	MIL-C-85806
Control Radio Set	C-10319A	MIL-C-85666
Receiver-Transmitter	RT-1250A/ARC	MIL-R-85664
	RT-1498/ARC	MIL-R-85664
Mounting, Electrical Equipment	MT-4609/ARC-159(V)	MIL-R-8177B

6.6 Subject term (key word) listing.

AN/ARC-182(V) Radio Set
Cipher Text
COMSEC
HAVE QUICK
Plain Text
Switching Unit
TSEC/KY-58
Wideband Audio

Preparing Activity:
Navy AS
(Project 5821-N196)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commander
Naval Air Systems Command (AIR 51122)
Washington, DC 20361

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:51Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:51Z
Check the source to verify that this is the current version before use.